ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · A6 · HUMANS AND ENVIRONMENT

# Helping or harming?

★☆☆
Grade 4

**Learning goals**

- Sort human actions.
- Use reduce, reuse and recycle.

## 1 Sort the actions. [6]

SORT THESE

**using a cloth bag** **planting trees** **throwing plastic in a river** **leaving lights on** **cleaning a beach** **hunting wild birds**

| Helps the environment | Harms the environment |
|---|---|
| planting trees | throwing plastic in a river |
| using a cloth bag | hunting wild birds |
| cleaning a beach | leaving lights on |

## 2 Write reduce, reuse or recycle for each action. [4]

| Action | Reduce, reuse or recycle? |
|---|---|
| taking a water bottle from home instead of buying one | reduce |
| using an old tin as a pencil pot | reuse |
| taking newspapers to the paper collector (*bhangari*) | recycle |
| printing on both sides of the paper | reduce |

## 3 Fill in the gaps. [4]

WORD BANK

**deforestation** **extinct** **habitat** **pollution** **sanctuary** **recycling**

Cutting down forests is called deforestation. An animal that has died out everywhere is extinct. Harmful things in the air or water are called pollution. A protected forest for animals is a wildlife sanctuary.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · A6 · HUMANS AND ENVIRONMENT

# Problems and solutions

★★☆
Grade 4

**Learning goals**

- Explain human impacts on living things.
- Suggest solutions.

**1** **For each problem, explain how it harms animals or plants and suggest a solution.** [6]

| Problem | How it harms | Solution |
|---|---|---|
| forests cut down for farms | animals lose food and homes | protect forests; plant new trees; farm without cutting more forest |
| factories pour dirty water into rivers | fish and plants die | clean the water before it leaves the factory; check with laws |
| too many young fish are caught | not enough fish grow up to breed | use bigger net holes; stop fishing in the breeding season |

**2** **Long ago, people in Bangladesh carried shopping in jute bags and baskets. Today many use plastic. How has this change affected the environment?** [2]

Plastic does not rot, so it blocks drains and rivers, causes floods and harms animals; jute bags rot away and can be reused many times.

**3** **Why is it important to tell others about caring for the environment?** [2]

Many people acting together make a much bigger difference than one person.

ANSWER KEY

INVESTIGATE · A6 · HUMANS AND ENVIRONMENT

# Our school litter survey

★★☆ Grade 4

**Learning goals**

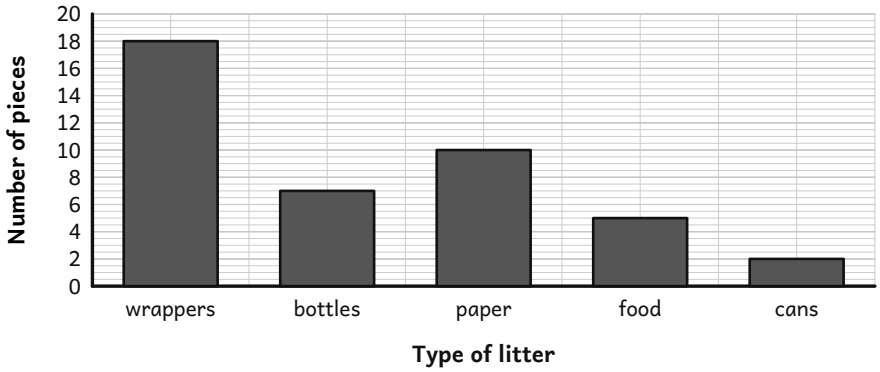
- Collect and tally data.
- Draw a bar chart.
- Plan action from evidence.

Class 4A wore gloves and collected litter from the school playground at the end of one day. They sorted it and counted each type.

**1** **Complete the tally chart.** [2]

| Type of litter | Tally | Total |
|---|---|---|
| sweet wrappers | IIII IIII IIII III | 18 |
| plastic bottles | IIII II | 7 |
| paper | IIII IIII | 10 |
| food waste | IIII | 5 |
| cans | II | 2 |

**2** **Draw a bar chart.** [3]




**3** **Which type of litter was most common? Suggest why.** [2]

Sweet wrappers; children eat snacks at break time and drop the wrappers.

**4** **Use the evidence to plan two actions for the school.** [2]

e.g. put bins next to the snack shop; sell snacks without wrappers; make a poster; have a weekly clean-up team.

**5** **Why did they wear gloves?** [1]

Litter can be dirty or sharp and have germs.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · A6 · HUMANS AND ENVIRONMENT

# The golden fibre: can jute beat plastic?

★★★ Grade 4

**Learning goals**

- Compare materials for the environment.
- Use numbers to compare.
- Weigh up different views.

Science · History · Economics · Design · Environment

**Jute** is a plant that grows well in the warm, wet fields of Bangladesh. Its long stems give a strong fibre that is made into bags, sacks, rope and mats. It was once so valuable that it was called the **golden fibre**, and Bangladesh is still one of the world's biggest growers.

Plastic bags are cheap and light, but they can last for hundreds of years and block drains and rivers. Jute bags rot away naturally in a few months, and jute plants take in a lot of carbon dioxide as they grow. A Bangladeshi scientist, Dr Mubarak Ahmad Khan, has developed a bag made from jute that looks like plastic, called the **Sonali bag**.

Based on information about jute in Bangladesh and the Sonali bag. Numbers below are rounded examples.

**1** **Compare a plastic bag and a jute bag.** [4]

| | Plastic bag | Jute bag |
|---|---|---|
| made from | oil (fossil fuel) | a plant that can be regrown |
| time to break down | hundreds of years | a few months |
| can it be reused many times? | a few times | yes, many times |
| cost | very cheap | costs more |

**2** **Maths. A family uses 5 plastic bags a week. How many is that in a year (52 weeks)? How many could one jute bag replace?** [2]

5 × 52 = 260 plastic bags a year; one strong jute bag used every week could replace all 260.

**3** **What might each person think about using jute bags?** [3]

| Person | View |
|---|---|
| a jute farmer | More jute bags means more work and money for farmers. |
| a shopkeeper | Jute bags cost more; I need cheap bags for customers. |
| a fish in the river | Please, no more plastic in my home! |

**4** **Design a campaign to get your school to use jute or cloth bags. Write a slogan and two ideas.** [2]

e.g. "Go golden with jute!"; give each learner a jute bag; a no-plastic snack day; a poster competition.

**ANSWER KEY**

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REVIEW AND CHALLENGE · A6 · HUMANS AND ENVIRONMENT

# Show what you know

★★★
Grade 4

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

## 1 Find the key words.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Y | U | R | E | C | Y | C | L | E | Y | K |
| R | E | D | U | C | E | B | F | F | V | J |
| I | Y | B | F | G | J | W | K | H | O | E |
| R | E | U | S | E | A | Y | L | M | C | R |
| T | V | P | L | A | S | T | I | C | R | E |
| R | O | D | Y | G | Y | K | T | Y | K | W |
| J | E | X | T | I | N | C | T | J | C | T |
| H | A | B | I | T | A | T | E | U | H | D |
| L | D | F | D | N | R | H | R | T | H | C |
| I | K | F | O | R | E | S | T | E | D | V |
| P | G | Y | J | U | M | I | H | S | S | E |

**extinct** **forest** **habitat** **jute** **litter** **plastic** **recycle** **reduce** **reuse**

## 2 Which action reduces waste? [1]

- ○ throwing old toys away
- ○ buying drinks in small plastic bottles
- ● **taking a cloth bag to the shop**
- ○ burning rubbish

## 3 An animal that is in danger of dying out is... [1]

- ○ extinct
- ● **endangered**
- ○ migrating
- ○ a producer

## 4 Draw a poster to stop people throwing plastic into rivers. [3]

My poster

A picture and message, e.g. "Rivers are not bins! Plastic kills fish."

**5** **Challenge. Give one way people have helped an ecosystem in Bangladesh and one way people have harmed one. Explain both.** [3]

Helped: protecting the Sundarbans as a sanctuary lets tigers and deer survive. Harmed: polluting the Buriganga killed fish and made the water unsafe.